Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	1 July 2026
Project Name	OptiCrop
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Home Page	USN-1	As a user, I can access the OptiCrop home page to learn about the system and its features.	2	High	**Mehar Gannamani**
Sprint-1	About OptiCrop	USN-2	As a user, I can read information about OptiCrop and understand its purpose and benefits.	2	Medium	**Ketha Mahidar**
Sprint-1	Feature Overview	USN-3	As a user, I can view the key features such as Smart Crop Prediction, Weather Analysis, and Higher Crop Yield.	2	High	**Mehar Gannamani**
Sprint-1	Navigation	USN-4	As a user, I can navigate easily between the Home page and Crop	2	High	**Ketha Mahidar**

			Recommendation page.			
Sprint -2	Crop Recommendation	USN-5	As a user, I can enter Nitrogen (N), Phosphorous (P), Potassium (K), Temperature, Humidity, pH, and Rainfall values.	3	High	**Mehar Gannamani**
Sprint -2	Input Validation	USN-6	As a user, I receive validation messages when invalid or incomplete values are entered.	2	High	**Ketha Mahidar**
Sprint -2	AI Prediction	USN-7	As a user, I can submit the input values and receive the most suitable crop recommendation using the Random Forest model.	5	High	**Mehar Gannamani**
Sprint -2	Prediction Status	USN-8	As a user, I can view the prediction completion status, AI model name, and prediction accuracy.	3	High	**Ketha Mahidar**
Sprint -3	Crop Recommendation Result	USN-9	As a user, I can view the recommended crop along with its image and AI confidence score.	3	High	**Mehar Gannamani**
Sprint -3	Crop Information	USN-10	As a user, I can read details about the recommended crop based on my	2	Medium	**Ketha Mahidar**

			soil and environmental conditions.			
Sprint -3	Growing Tips	USN-11	As a user, I can view general growing tips for the recommended crop to improve productivity.	2	High	**Mehar Gannamani**
Sprint -3	Weather Analysis	USN-12	As a user, I can understand how weather and environmental parameters influence crop prediction.	3	Medium	**Ketha Mahidar**
Sprint -4	Predict Again	USN-13	As a user, I can predict another crop without restarting the application.	2	High	**Mehar Gannamani**
Sprint -4	Home Navigation	USN-14	As a user, I can return to the home page after viewing the prediction result.	1	High	**Ketha Mahidar**
Sprint -4	Responsive Design	USN-15	As a user, I can use the application smoothly on both mobile and desktop devices.	2	Medium	**Mehar Gannamani**
Sprint -4	Machine Learning Integration	USN-16	As a system, I use the trained Random Forest model to provide accurate crop recommendations based on user inputs.	5	High	**Ketha Mahidar**, **Mehar Gannamani**

Project Tracker, Velocity s Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	8	3 Days	24 Jun 2026	26 Jun 2026	8	26 Jun 2026
Sprint-2	13	3 Days	27 Jun 2026	29 Jun 2026	13	29 Jun 2026
Sprint-3	10	2 Days	30 Jun 2026	01 Jul 2026	10	01 Jul 2026
Sprint-4	10	3 Days	02 Jul 2026	04 Jul 2026	10	04 Jul 2026

Velocity:

Velocity = Total Story Points Completed ÷ Number of Sprints

= 41 ÷ 4

= 10.25 Story Points per Sprint

Average Velocity

Total project duration = 11 Days

Average Velocity per Day

= 41 ÷ 11

= 3.73 Story Points per Day

Burndown Chart:

OptiCrop Sprint Burndown Chart

Remaining story points after each sprint.